

Software Requirements Specification

TruLabel

An Agentic AI System for Verifying Product Marketing Claims Against Credible Evidence

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1. Introduction

1.1 Purpose

This document specifies the software requirements for TruLabel, an agentic AI system designed to help consumers verify the authenticity of product marketing claims (e.g., “healthy,” “100% natural,” “no added sugar”) against credible, independently verifiable sources. This SRS is intended for the project supervisor, evaluators, and the developer (the author) as a reference for scope, functional behaviour, and design constraints throughout development.

1.2 Scope

TruLabel allows a user to search for, or submit details of, a packaged consumer product (initially scoped to packaged food and health/wellness products) and receive an automatically generated report that:

- Extracts specific, checkable marketing claims from the product's advertised description or label text.
- Retrieves relevant evidence for each claim from open nutrition/ingredient data sources and a curated set of regulatory reference material.
- Weighs evidence by source credibility.
- Produces a plain-language verdict and an overall “Marketing Accuracy Score” for the product.

The system is a decision-support tool for consumers. It does not issue legal findings, does not replace regulatory action by bodies such as FSSAI or the CCPA, and its verdicts are advisory, not authoritative.

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
Agent	An autonomous software component with a defined role (e.g., extraction, retrieval) that performs a task and passes structured output to the next stage of the pipeline.
Claim	A specific, checkable assertion made in a product's marketing material (e.g., “no added sugar”).
FSSAI	Food Safety and Standards Authority of India — the statutory food regulator.
CCPA	Central Consumer Protection Authority — the body empowered to act against misleading advertisements in India.
MAS	Marketing Accuracy Score — the composite output score produced by TruLabel for a product.
OFF	Open Food Facts — an open, crowd-sourced food product database used as an evidence source.

1.4 References

- Open Food Facts API documentation — openfoodfacts.github.io/openfoodfacts-server/api/
- FSSAI (Food Safety and Standards (Advertising and Claims) Regulations, 2018)
- CCPA Guidelines for Prevention of Misleading Advertisements and Endorsements, 2022
- IEEE Std 830-1998 — Recommended Practice for Software Requirements Specifications (structural reference for this document)

1.5 Overview

Section 2 describes the product at a high level. Section 3 lists specific functional and non-functional requirements. Section 4 outlines system models referenced elsewhere in the project's UML and architecture documents. Section 5 lists assumptions, dependencies, and open issues.

2. Overall Description

2.1 Product Perspective

TruLabel is a new, standalone, self-contained system. It is a full-stack web application comprising a user-facing frontend, a backend orchestration service coordinating a pipeline of specialised agents, a persistent database, and integrations with an external open data source (Open Food Facts) and a locally curated regulatory reference dataset. It does not extend or replace any existing university or commercial system.

2.2 Product Functions

At a high level, TruLabel performs the following functions:

- Accepts a product search query or label submission (text or image) from the user.
- Extracts discrete marketing claims from the product's description/label using an LLM-based extraction agent.
- Retrieves ingredient and nutrition ground-truth data for the product from Open Food Facts where available.
- Cross-references each claim against ground-truth data and against the curated regulatory rule set (FSSAI claims regulations, known CCPA rulings).
- Scores each claim and produces a weighted overall Marketing Accuracy Score.
- Presents a plain-language report to the user, including a per-claim breakdown and supporting evidence citations.
- Stores search history and generated reports for later retrieval.

2.3 User Characteristics

User Type	Description
General Consumer	Everyday user with no technical background, searching for a product before purchase. Requires a simple search interface and plain-language output.
Evaluator / Supervisor	Reviews the system for academic assessment; may need visibility into agent reasoning steps and evidence sources for demonstration purposes.

2.4 Constraints

- Ground-truth ingredient/nutrition data is limited to what is available in Open Food Facts, which has partial and crowd-sourced India coverage; products absent from OFF cannot be fully verified against nutrition data.
- No live, structured API exists for FSSAI product-level claims data or CCPA rulings; the regulatory reference set used by TruLabel is a manually curated, static dataset rather than a live feed, and must be clearly scoped as such.
- The system's verdicts are advisory and must be presented with a visible disclaimer; TruLabel does not perform legal adjudication.

- Initial scope is limited to packaged food and health/wellness products to keep the evidence sources tractable within a single semester.

2.5 Assumptions and Dependencies

- Open Food Facts API remains publicly accessible and free to use for the duration of the project.
- An LLM API (e.g., via the Anthropic or an equivalent provider) is available for the extraction, scoring, and synthesis agents.
- The curated regulatory dataset (FSSAI claims rules + a sample of CCPA rulings) will be compiled manually by the developer as part of project setup and is assumed sufficient for demonstration and evaluation purposes, not for production-grade legal coverage.

3. Specific Requirements

3.1 Functional Requirements

ID	Requirement	Priority
FR-1	The system shall allow a user to search for a product by name or brand.	High
FR-2	The system shall allow a user to submit a product label via image upload for OCR-based text extraction.	Medium
FR-3	The system shall extract a list of discrete, checkable marketing claims from the product's description or label text.	High
FR-4	The system shall retrieve ingredient and nutrition data for the identified product from Open Food Facts, where available.	High
FR-5	The system shall cross-reference each extracted claim against retrieved ingredient/nutrition data and the curated regulatory rule set.	High
FR-6	The system shall assign a credibility weight to each evidence source used in generating a verdict.	Medium
FR-7	The system shall compute a per-claim verdict and an overall Marketing Accuracy Score (MAS) for the product.	High
FR-8	The system shall present a plain-language report to the user, including per-claim verdicts, the MAS, and cited evidence.	High
FR-9	The system shall persist past searches and generated reports for a given user session.	Low
FR-10	The system shall display a disclaimer clarifying that verdicts are advisory and not a legal or regulatory finding.	High
FR-11	The system shall allow a user to select two or more products and view their verdicts side by side via a comparison agent.	Medium
FR-12	The system shall allow a user to request alternative products in the same category as a selected product.	Medium
FR-13	The system shall retrieve candidate alternative products in the same category from Open Food Facts and rank them using a lightweight, non-LLM heuristic (e.g., sugar content, additive count, available Nutri-Score) prior to full verification.	Medium
FR-14	The system shall run the full verification pipeline only on the top-ranked shortlist of candidate alternatives, and shall present a recommended alternative with a plain-language justification via a recommendation agent.	Medium

3.2 Non-Functional Requirements

ID	Category	Requirement
NFR-1	Performance	A full claim-verification report shall be generated within a target of 15–30

		seconds for a product with available Open Food Facts data.
NFR-2	Usability	The interface shall be usable by a non-technical consumer without instructions, using plain-language output rather than raw agent/technical logs.
NFR-3	Reliability	If an evidence source (e.g., Open Food Facts) is unavailable, the system shall degrade gracefully and clearly indicate reduced confidence rather than fail silently.
NFR-4	Maintainability	Each agent in the pipeline shall be implemented as an independent, replaceable module with a clearly defined input/output contract.
NFR-5	Security	User-submitted images and search history shall not be exposed to other users; API keys and credentials shall not be hard-coded in source control.
NFR-6	Transparency	Every verdict shall be traceable to the specific evidence and source used to generate it.

3.3 External Interface Requirements

3.3.1 User Interfaces

A web-based search interface (desktop and mobile-responsive) allowing text search or image upload, and a results view presenting the MAS and per-claim breakdown.

3.3.2 Software Interfaces

- Open Food Facts REST API (product ingredient/nutrition lookup).
- An LLM provider API for claim extraction, evidence reasoning, and verdict synthesis.
- OCR service/library for label image text extraction (if FR-2 is implemented).

3.3.3 Communication Interfaces

Standard HTTPS REST calls between the frontend, backend orchestrator, and external APIs.

4. System Models (Summary)

Detailed UML diagrams (use case, class, sequence, and activity) and the full system architecture diagram are maintained as separate project deliverables. At a system level, TruLabel follows a five-stage core verification pipeline coordinated by a backend orchestrator:

- Claim Extraction Agent → Evidence Retrieval Agent → Ingredient/Nutrition Reality-Check Agent → Credibility-Weighting Agent → Verdict Synthesis Agent.

Each stage consumes structured output from the previous stage and passes structured output forward, allowing agents to be tested, replaced, or extended independently.

Two additional modes extend this core pipeline (FR-11 to FR-14):

4.1 Compare Mode

Two or more user-selected products are each run independently through the full core pipeline. A Comparison Agent then aligns the resulting verdicts and evidence side by side for the user.

4.2 Recommend Mode

Given a selected product, a Category Candidate Search Agent retrieves other products in the same Open Food Facts category. A Quick Heuristic Filter (non-LLM, based on raw nutrition/ingredient data such as sugar content, additive count, or available Nutri-Score) narrows this list to a small shortlist, which alone is passed through the full core pipeline. A Recommendation Agent then selects and justifies the best-performing alternative from the shortlist. This two-stage filter-then-verify design

avoids running the full (LLM-dependent) verification pipeline on every product in a category, which would be computationally and financially impractical for categories with large numbers of listings.

5. Assumptions, Dependencies, and Open Issues

- Final choice of agent-orchestration framework (e.g., LangGraph, CrewAI, or a custom sequential pipeline) is to be finalised during system design.
- The size and coverage of the manually curated CCPA/regulatory rulings dataset is to be finalised before the mid-term review.
- Whether OCR-based label submission (FR-2) is included in the MVP or deferred to a later iteration depends on time available after core pipeline (FR-1, FR-3–FR-8) is stable.
- The exact heuristic used in the Quick Heuristic Filter (FR-13) — e.g., weighting of sugar content vs. additive count vs. Nutri-Score — is to be finalised during system design.
- Open Food Facts category tagging consistency for India-specific products should be spot-checked early, as inconsistent categorisation would directly affect the quality of Recommend Mode (FR-12–FR-14).

End of Document — Draft v0.1